

Campfire-1

Question : Alonzo Spotted Weird files on his computer and informed the newly assembled SOC Team. Assessing the situation it is believed a Kerberoasting attack may have occurred in the network. It is your job to confirm the findings by analyzing the provided evidence.

You are provided with:

- 1- Security Logs from the Domain Controller
- 2- PowerShell-Operational Logs from the affected workstation
- 3- Prefetch Files from the affected workstation

This scenario has undergone a Kerberoasting Attack

Whats is Kerberos Authentication

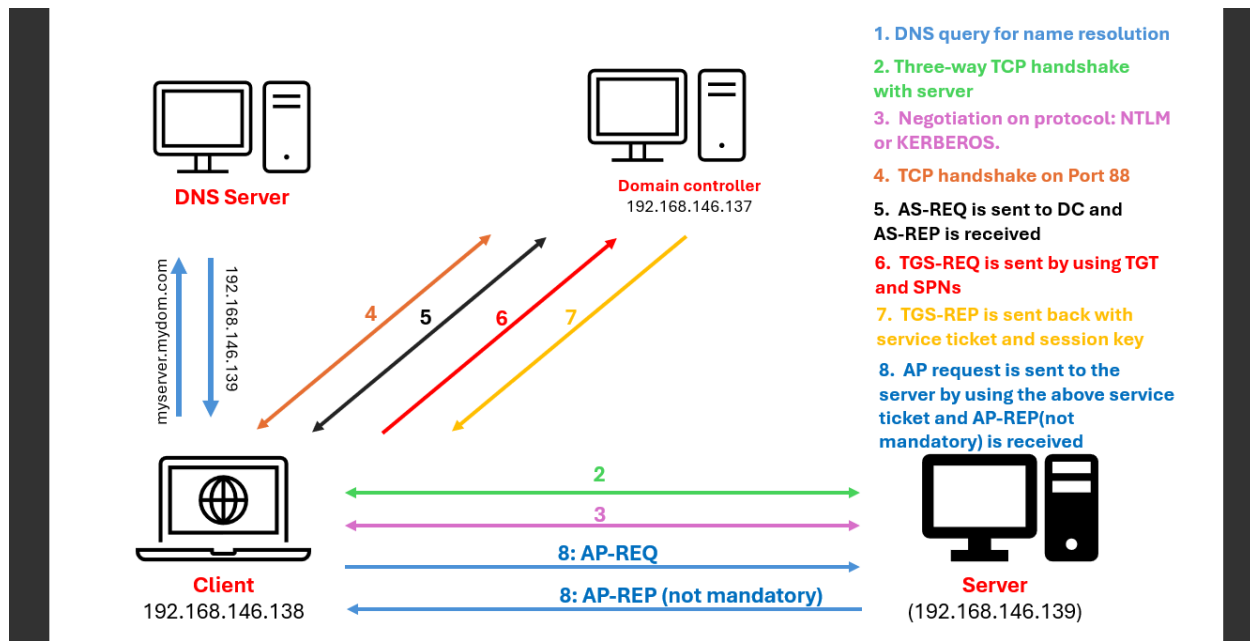
Kerberos is a authentication protocol used for authentication in WindowsActive Directory, They actually work like the user no need to send the password packets in network repeatedly for security purpose.

The core idea: prove who you are once, then use tickets instead of repeatedly sending your password

It has three components

- Clients - users/computers
- KDC (Key Distribution centre) runs in DC. It has 2 parts Authentication server and TGS Ticket Granting Service
- Services

<https://medium.com/@SeverSerenity/kerberos-authentication-process-b9c7db481c56>



Kerberoasting Attack is that the Ticket is encrypted with a key through the Server Accounts password hash, so now the attacker brute -forces the password if the password is weak he could crack it. Now he can access the Service account anywhere it may enables file shares, databases, applications, sometimes even RDP.

Before stepping into the Challenge look what file you got to investigate

1.SECURITY-DC.evtx

2.Powershell-Operational.evtx

3.Z:\htb\Sherlocks\Campfire-1\Triage\Workstation\2024-05-21T033012_triage_asset\C\Windows\prefetch

in this directory we can find 212 prefetch file, we now have to convert the .pf file to .csv format for investigation we can do this through KAPE "Eric Zimmerman Tool" it has PECmd.exe that does the conversion of the .pf file to .csv file

In the cmd prompt terminal paste this command

```
PECmd.exe -d "Z:\htb\Sherlocks\Campfire-1\Triage\Workstation\2024-05-21T033012_triage_asset\C\Windows\prefetch" --csv "Z:\htb\Sherlocks\Campfire-1" --csvf analysis.csv
```

Name	Date modified	Type	Size
 Triage	5/20/2024 10:03 PM	File folder	
 analysis.csv	9/9/2026 10:56 PM	Microsoft Excel C...	2,993 KB
 analysis_Timeline.csv	9/9/2026 10:56 PM	Microsoft Excel C...	78 KB
 campfire-1.zip	9/9/2026 10:07 PM	Compressed (zipp...	3,949 KB

Ignore the analysis_Timeline.csv just look on to anlalysis.csv file alone.

we get the analysis.csv file now we have to investigate that, now lets move on to analysis phase.

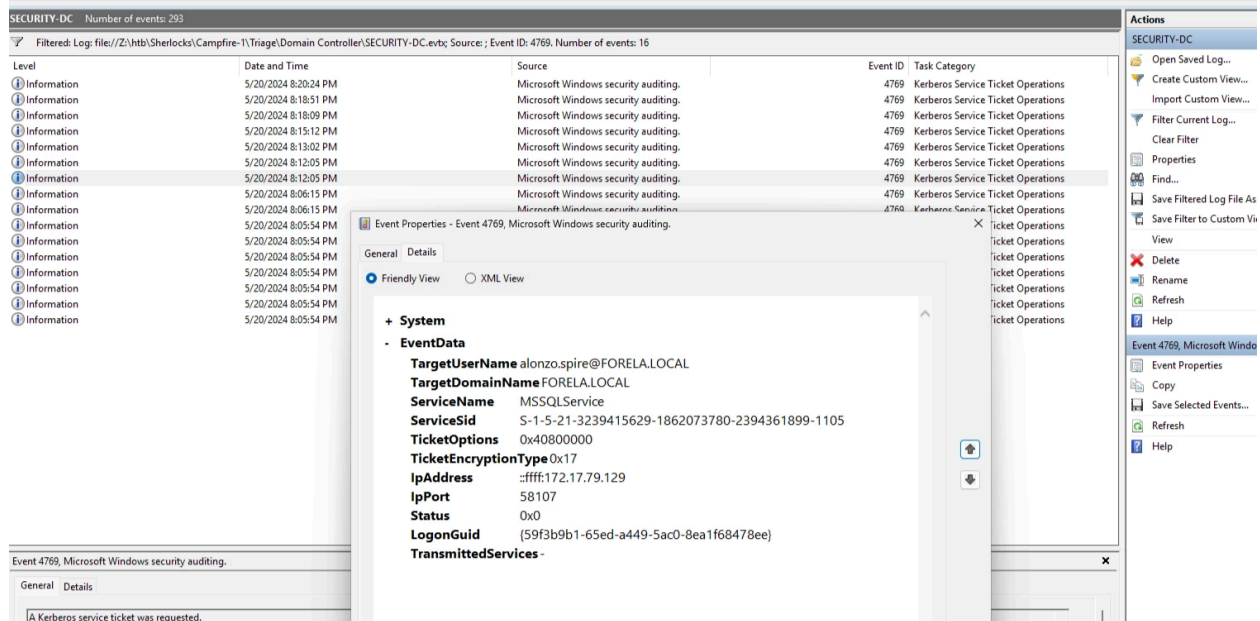
1.Analyzing Do main Controller Security Logs, can you confirm the UTC date & time when the kerberoasting activity occurred?

HINT: In Security Logs, **Filter for Event ID 4769**. Now Look for any event where the service name is NOT(krbtgt or ends with (*Fore.gDC01*)). The ticket type should be 0x17 which is for RC4 type encryption. The failure code should be 0x0. The event that matches all the above conditions is the event detailing information about the kerberoasting attack activity

krbtgt is a in - built active directory account used by kerbose for ticket infrastructure

A computer account in Active Directory normally ends with a **dollar sign \$**

Event ID 4769 means **A Kerberos service ticket was requested**



2024-05-21 03:18:09

2.What is the Service Name that was targeted?

MSSQLService

3.It is really important to identify the Workstation from which this activity occurred. What is the IP Address of the workstation?

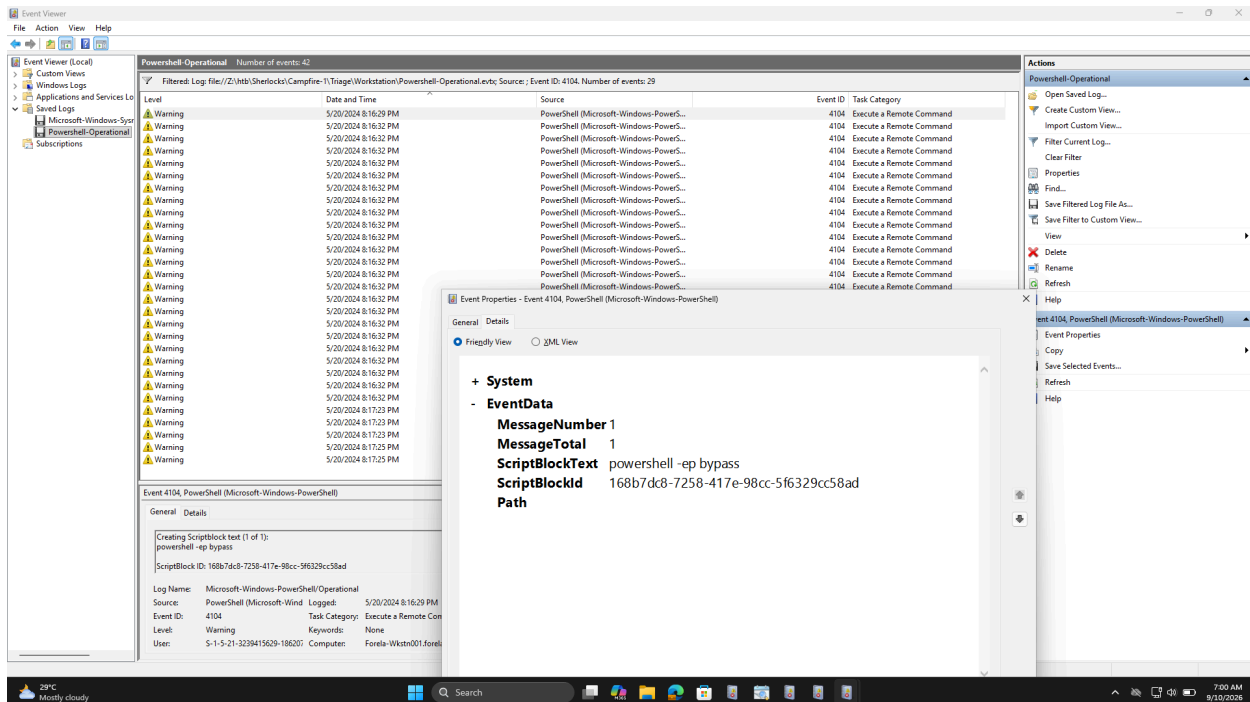
172.17.79.129

4.Now that we have identified the workstation, a triage including PowerShell logs and Prefetch files are provided to you for some deeper insights so we can understand how this activity occurred on the endpoint. What is the name of the file used to Enumerate Active directory objects and possibly find Kerberoastable accounts in the network?

HINT : Use PowerShell logs and filter for event ID 4104. We can see all the contents of the script executed and its name as well.

Now we have to look through Powershell event logs, a separate event log other than the SECURITY - DC.evtx

Event ID 4104 means **PowerShell Script Block Logging recorded PowerShell code that was executed**



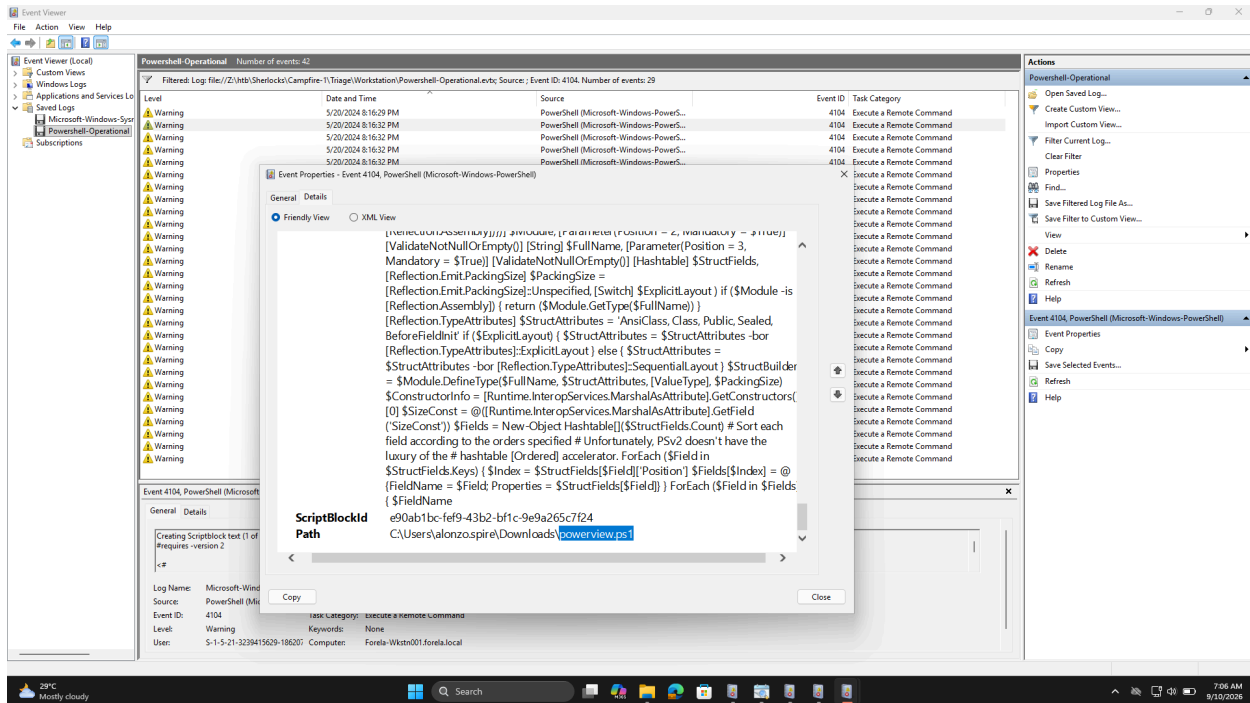
You see the first event of the powershell cmd executed the attacker used the cmd "powershell -ep bypass ". It starts a **new PowerShell process** with its execution policy set to **Bypass for that process**.

what is Execution policy in Powershell - it can restrict how powershell scripts are executed

Example policies

- 1.Restricted
- 2.AllSigned
- 3.RemoteSigned
- 4.Unrestricted
- 5.Bypass

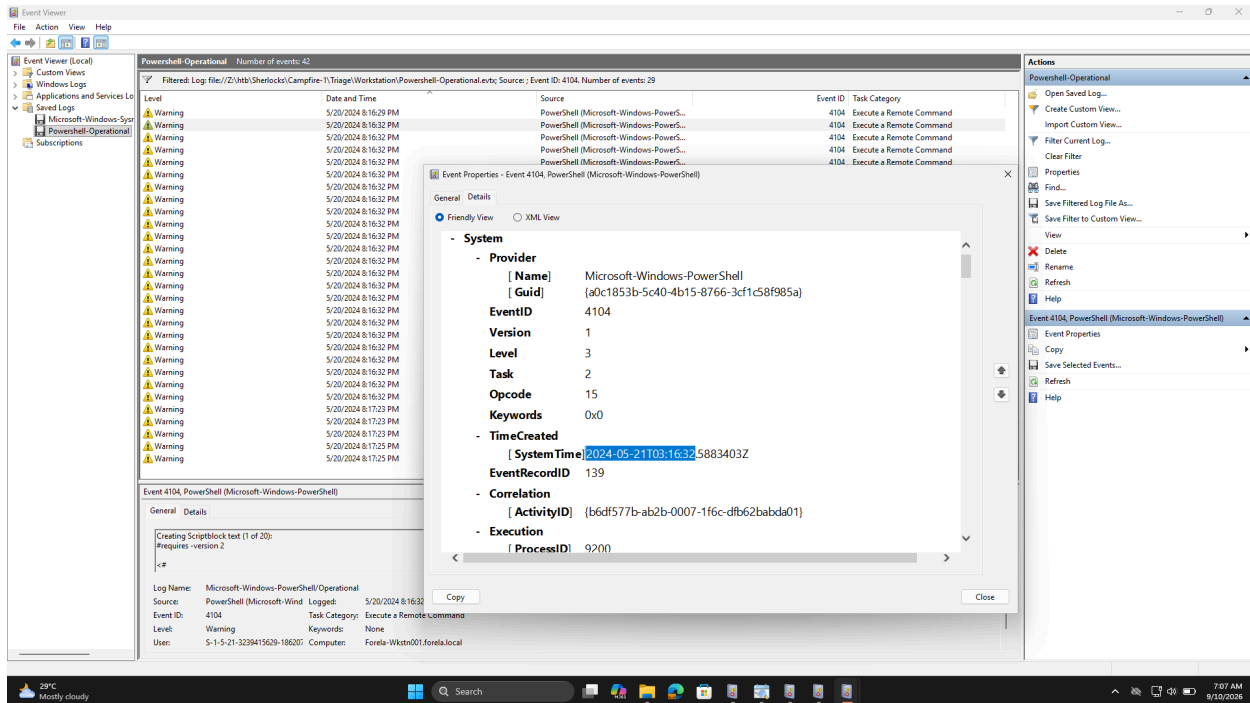
Now with this bypass cmd enforces the powershell scripts to run without any policy restrictions



powerview.ps1

5. When was this script executed? (UTC)

In the Powershell.evtx look for the Date and time make sure it is in order. The second event click it and go watch the details under the System we can see the SystemTime 2024-05-21T03:16:32



2024-05-21 03:16:32

6.What is the full path of the tool used to perform the actual kerberoasting attack?

From the above solve and reconned phase we got to know that at this "2024-05-21 03:18:09"

day the Kerberoasting attack had done. This is the time when the Kerbose ticket was request so search the near by timings

Click the filter icon in the Last Run column and set the Date

Modif...	Source Access...	Executable Name	Run Count	Hash	Size	Version	Last Run	Pre
	-		-		-		-	-
i-21 0...	2026-09-10 05...	SHELLEXPERIENCEHOST.EXE	4	655318BF	186616	Windows ...	2024-05-21 03:14:08	202...
i-21 0...	2026-09-10 05...	MICROSOFTEDGEUPDATE.EXE	152	7A595326	55238	Windows ...	2024-05-21 03:14:20	202...
i-21 0...	2026-09-10 05...	DLLHOST.EXE	28	47BE07DC	58074	Windows ...	2024-05-21 03:15:03	202...
i-21 0...	2026-09-10 05...	DLLHOST.EXE	41	1BAE068B	19546	Windows ...	2024-05-21 03:15:03	202...
i-21 0...	2026-09-10 05...	SVCHOST.EXE	147	9A28EB78	21822	Windows ...	2024-05-21 03:15:07	202...
i-21 0...	2026-09-10 05...	MSEDGE.EXE	11	37D25F9E	23892	Windows ...	2024-05-21 03:15:42	202...
i-21 0...	2026-09-10 05...	MSEDGE.EXE	36	37D25F9A	466380	Windows ...	2024-05-21 03:15:42	202...
i-21 0...	2026-09-10 05...	RUNTIMEBROKER.EXE	1	5A3B22F7	19784	Windows ...	2024-05-21 03:15:43	
i-21 0...	2026-09-10 05...	IDENTITY_HELPER.EXE	1	4E2A63C3	75952	Windows ...	2024-05-21 03:15:43	
i-21 0...	2026-09-10 05...	ELEVATION_SERVICE.EXE	1	479FE2CB	21352	Windows ...	2024-05-21 03:15:43	
i-21 0...	2026-09-10 05...	MSEDGE.EXE	51	37D25F9C	157148	Windows ...	2024-05-21 03:15:43	202...
i-21 0...	2026-09-10 05...	MSEDGE.EXE	13	37D25F9D	175394	Windows ...	2024-05-21 03:15:43	202...
i-21 0...	2026-09-10 05...	BACKGROUNDTASKHOST.EXE	11	F8B2DD01	84464	Windows ...	2024-05-21 03:15:43	202...
i-21 0...	2026-09-10 05...	CONHOST.EXE	14	C6456F8	38326	Windows ...	2024-05-21 03:16:00	202...
i-21 0...	2026-09-10 05...	POWERSHELL.EXE	2	CA1AE517	151870	Windows ...	2024-05-21 03:16:29	202...
i-21 0...	2026-09-10 05...	SPPSVC.EXE	123	96070FE0	52212	Windows ...	2024-05-21 03:16:50	202...
i-21 0...	2026-09-10 05...	MPCMDRUN.EXE	4	D65F9B9	26756	Windows ...	2024-05-21 03:16:50	202...
i-21 0...	2026-09-10 05...	RUBEUS.EXE	1	5873E24B	86612	Windows ...	2024-05-21 03:18:08	
i-21 0...	2026-09-10 05...	SVCHOST.EXE	38	952637C2	24670	Windows ...	2024-05-21 03:22:18	202...
i-21 0...	2026-09-10 05...	MMC.EXE	14	1EE19326	349966	Windows ...	2024-05-21 03:22:52	202...
i-21 0...	2026-09-10 05...	MSEDGE.EXE	125	37D25FA2	55748	Windows ...	2024-05-21 03:25:43	202...
i-21 0...	2026-09-10 05...	SEARCHFILTERHOST.EXE	54	44162447	16414	Windows ...	2024-05-21 03:26:51	202...
i-21 0...	2026-09-10 05...	SEARCHPROTOCOLHOST.EXE	44	69C456C3	20204	Windows ...	2024-05-21 03:26:51	202...
i-21 0...	2026-09-10 05...	TRUSTEDINSTALLER.EXE	2	766EFF52	17414	Windows ...	2024-05-21 03:27:06	202...
i-21 0...	2026-09-10 05...	TIWORKER.EXE	2	F8D79BD6	131950	Windows ...	2024-05-21 03:27:06	202...
i-21 0...	2026-09-10 05...	TASKHOSTW.EXE	258	2E5D4B75	113712	Windows ...	2024-05-21 03:27:06	202...
i-21 0...	2026-09-10 05...	SVCHOST.EXE	1	F952D9A9	43346	Windows ...	2024-05-21 03:27:07	
i-21 0...	2026-09-10 05...	DLLHOST.EXE	103	7D5CE0CA	15772	Windows ...	2024-05-21 03:28:11	202...
i-21 0...	2026-09-10 05...	WINRAR.EXE	18	BA8CDB31	425610	Windows ...	2024-05-21 03:28:14	202...
i-21 0...	2026-09-10 05...	SMARTSCREEN.EXE	80	EACC1250	60000	Windows ...	2024-05-21 03:28:14	202...
i-21 0...	2026-09-10 05...	FILECOAUTH.EXE	4	88F71F64	48786	Windows ...	2024-05-21 03:28:17	202...
i-21 0...	2026-09-10 05...	SVCHOST.EXE	100	85265587	26550	Windows ...	2024-05-21 03:28:18	202...

RUBEUS.EXE Bingooo

Rubeus.exe is an open-source C# tool used to interact with and abuse the Kerberos authentication protocol in Windows Active Directory environments, so it is malicious Lock in.

In the .csv file go to the Files Loaded column click the RUBEUS.exe row

c:\USERS\ALONZO.SPIRE\DOWNLOADS\RUBEUS.EXE

7.When was the tool executed to dump credentials? (UTC)

See the last run column

Timeline Explorer v2026.5.0

FileToolsTabsViewHelp

analysis.csv

Drag a column header here to group by that column

Enter text to search...

Find

	Volume1Seri...	Source Created	Source Modif...	Source Access...	Executable Name	Run Count	Hash	Size	Version	Last Run	Previous Ru...	Previous Ru...	Previous
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	SHELLEXPERIENCEHOST.EXE	4	6553188F	186616	Windows ...	2024-05-21 03:14:08	2024-04-26 ...	2024-03-26 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	MICROSOFTEDGEUPDATE.EXE	152	7A595326	55238	Windows ...	2024-05-21 03:14:20	2024-05-21 ...	2024-05-14 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	DLLHOST.EXE	28	478E07DC	58074	Windows ...	2024-05-21 03:15:03	2024-03-26 ...	2024-03-19 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	DLLHOST.EXE	41	1BAE0688	19546	Windows ...	2024-05-21 03:15:03	2024-04-26 ...	2024-03-26 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	SVCHOST.EXE	147	9A28EB78	21822	Windows ...	2024-05-21 03:15:07	2024-05-21 ...	2024-05-14 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	MSEDGE.EXE	11	37025F9E	23892	Windows ...	2024-05-21 03:15:42	2024-04-26 ...	2024-03-26 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	MSEDGE.EXE	36	37025F9A	466380	Windows ...	2024-05-21 03:15:42	2024-05-21 ...	2024-04-26 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	RUNTIMEBROKER.EXE	1	5A3B22F7	19784	Windows ...	2024-05-21 03:15:43			
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	IDENTITY_HELPER.EXE	1	4E2A63C3	75952	Windows ...	2024-05-21 03:15:43			
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	ELEVATION_SERVICE.EXE	1	479FE2C8	21352	Windows ...	2024-05-21 03:15:43			
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	MSEDGE.EXE	51	37025F9C	157148	Windows ...	2024-05-21 03:15:43	2024-04-26 ...	2024-03-26 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	MSEDGE.EXE	13	37025F9D	175394	Windows ...	2024-05-21 03:15:43	2024-04-26 ...	2024-03-26 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	BACKGROUNDTASKHOST.EXE	11	F8B2D081	84464	Windows ...	2024-05-21 03:15:43	2024-05-21 ...	2024-04-26 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	CONHOST.EXE	14	C6456FB	38326	Windows ...	2024-05-21 03:16:00	2024-05-21 ...	2024-05-21 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	POWERSHELL.EXE	2	C1A1A517	151870	Windows ...	2024-05-21 03:16:29	2024-05-21 ...		
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	SPPSVC.EXE	123	96070FE0	52212	Windows ...	2024-05-21 03:16:50	2024-05-21 ...	2024-05-14 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	NPCMDRUN.EXE	4	D65F989	26756	Windows ...	2024-05-21 03:16:50	2024-05-21 ...	2024-05-21 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	RUBEUS.EXE	1	5873E248	86612	Windows ...	2024-05-21 03:18:08			
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	SVCHOST.EXE	38	952637C2	24670	Windows ...	2024-05-21 03:22:18	2024-05-21 ...	2024-04-26 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	NMC.EXE	14	1EE19326	349966	Windows ...	2024-05-21 03:22:52	2024-03-19 ...	2024-03-14 ...	2023-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	MSEDGE.EXE	125	37025FA2	55748	Windows ...	2024-05-21 03:25:43	2024-05-21 ...	2024-04-26 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	SEARCHFILTERHOST.EXE	54	44162447	16414	Windows ...	2024-05-21 03:26:51	2024-05-21 ...	2024-05-14 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	SEARCHPROTOCOLHOST.EXE	44	69C456C3	20204	Windows ...	2024-05-21 03:26:51	2024-05-21 ...	2024-05-14 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	TRUSTEDINSTALLER.EXE	2	766EFF52	17414	Windows ...	2024-05-21 03:27:06	2024-05-21 ...		
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	TIWORKER.EXE	2	F8D79806	131950	Windows ...	2024-05-21 03:27:06	2024-05-21 ...		
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	TASKHOSTW.EXE	258	2E5D4B75	113712	Windows ...	2024-05-21 03:27:06	2024-05-21 ...	2024-05-21 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	SVCHOST.EXE	1	F95D9A9	43346	Windows ...	2024-05-21 03:27:07			
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	DLLHOST.EXE	103	7D5CE0CA	15772	Windows ...	2024-05-21 03:28:11	2024-05-21 ...	2024-05-21 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	WINRAR.EXE	18	BABCD831	425610	Windows ...	2024-05-21 03:28:14	2024-03-19 ...	2024-03-19 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	SMARTSCREEN.EXE	80	EACC1250	60000	Windows ...	2024-05-21 03:28:14	2024-05-21 ...	2024-05-21 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	FILECOAUTH.EXE	4	8BF71F64	48786	Windows ...	2024-05-21 03:28:17	2024-05-21 ...	2024-05-21 ...	2024-0...
rkstation\...		2026-09-10 05:10:56	2024-05-21 0...	2026-09-10 05...	EXPLORER.EXE	100	8533F88	65550	Windows ...	2024-05-21 03:30:10	2024-05-14 ...	2024-04-26 ...	2024-0...

Last Run

Is same day

2024-05-21 00:00:00

Edit Filter

Z:\HB\Sherlock\Campfire-TiAnalysis.csv

Total lines 207Visible lines 87Open files: 1

Search options

29°C

Mostly cloudy

Search

4.01

7:25 AM

9/10/2026

2024-05-21 03:18:08